

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

May 2012

EC209 Microprocessors & Microcontroller

Date: 15/05/2012, Tuesday

Time: 1:30 p.m. to 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Briefly give answers of the following questions: (each 3 marks) [15]

1. What is the need of ALE signal in 8085?
2. List the features of Microcontroller with respect to Microprocessor.
3. Differentiate the absolute and partial address decoding techniques.
4. If the memory-chip size is 2048×4 bits, how many chips are required to make up 16K-byte memory?
5. Explain how an instruction is decoded and executed in 8085?

Q - 2 (a) List the assembler directives in 8085 μ P. Give explanation of each with an example. [05]

(b) List the machine cycle required for execution of following instructions [03]

1. LHLD 2501H
2. LXI H, 2501H
3. LDA 2501H

(c) Give the significance of SIM and RIM instruction available in 8085. [02]

OR

Q - 2 (a) Draw the timing diagram for MOV A, M instruction. [04]

(b) Distinguish between synchronous and asynchronous data transfer. [03]

(c) Compare the following pairs of instructions with their operations, instruction byte, addressing modes, affected flags and the results. [03]

1. MVI A,00H and XRA A
2. SUB B and CMP B
3. JMP 2700H and PCHL

Q - 3 (a) Write an ALP to multiply the contents of memory location 2040 H to 2041 H using repetitive addition concept. Store the result at 2500H. [05]

(b) What do you mean by opcode and operands? [02]

(c) What will be the status of CY flag after execution of CMP L instruction? [03]

OR

Q - 3 (a) Write an ALP to convert 8 bit binary to ASCII code. Draw the flowchart of the logic program. [07]

(b) What is the function performed by timing and control unit in a microprocessor? [03]

SECTION – II

- Q - 4 (a) What are the advantages of an assembly language in comparison with high level languages? [02]
(b) List out different interrupts of 8085 & explain them briefly. [05]
(c) What is the major difference between PC and DPTR registers of 8051 microcontroller? [03]
- Q - 5 (a) Write a program in assembly language to find the square root of a number. (say, $X = \sqrt{N}$, where N is a number). [08]
(b) How the internal RAM of 8051 microcontroller is organized? [02]

OR

- Q - 5 (a) Write a program to generate a delay of 0.5 sec if the crystal frequency is 5 MHz. Also, give calculation of number of T-states required to generate the delay. [08]
(b) What are register banks in 8051 microcontroller? [02]
- Q - 6 Write short note on **any three** of the given topics (each 5 marks) [15]
(a) DMA controller with Block diagram
(b) 8255 PPI and its various modes
(c) Interfacing of Memory (2 KB RAM) with 8085 μ P (Use address range: A800H-AFFFFH)
(d) Timers and Counters in 8051 Microcontroller
(e) Static and Dynamic RAM